

Written Documentation

Architecture, wire contracts, setup, research and module guides

EdgePilot AI - Team 15

Documentation set, 23 August 2026

5 documents in this volume

docs/ - long-form documentation

Work package: shared; each subfolder belongs to the module it describes.

What this folder is

The written record: architecture, wire contracts, setup, research reasoning, and per-module guides. 00-PROJECT-RECORD.md at the root is the history; this folder is the reference.

What happened here

docs/deployment.md was removed. It described a deployment that does not exist - there is no public URL - and a document that describes a thing into existence is worse than an admission that the thing is missing. The honest version is section 6 of 00-PROJECT-RECORD.md.

docs/internal/README.md was updated to point at the current set.

One document in here is **known stale** and is listed as a gap rather than quietly patched: docs/internal/database.md line 147 still says users own "devices", and it does not mention shared_findings, session_id, warmup or privacy_class.

Files

Subfolder / file	What it is
local-model-setup.md	352 lines. Clean machine to measured benchmark: Docker, WSL2, Ollama, API keys, verification, troubleshooting, and the error-code table
benchmark/	The benchmark and provider layer guide, plus its research log
dashboard/	The dashboard module guide
internal/	Architecture, database, wire contract, team workflow, project plan
vision-benchmark/	Dataset card, workload spec, provider execution, test cases, evidence report

Connected folders

- / - README.md and CONTRIBUTING.md are the public-facing counterparts to this folder.
- src/modules/benchmark/ and src/modules/vision-benchmark/ - the code these documents describe.

Referenced from

These folders point here. Each link below resolves in both directions.

- tests/benchmark/

docs/benchmark/ - benchmark and provider layer documentation

Work package: AI / local model & DevOps - the benchmark and provider layer.

What this folder is

The branch guide for the measurement core: what is in it, what was decided, what evidence backs each decision, and what it deliberately does not do.

What happened here

These two documents are the reason the fix list in 00-PROJECT-RECORD.md could be written with reasoning rather than guesswork - the research log records why each decision was taken **and what evidence supports it**, including the decisions to *not* measure certain things.

The guide includes a section written for the integration work package, and a statement that there is no authenticated session and ownership is derived from `workloads.user_id`. That statement is now historically accurate rather than currently accurate: a session id exists (see `src/lib/`), though the ownership rule it describes survived on its own merits - a benchmark belongs to whoever owns the workload being benchmarked.

Files

File	What it is
README.md	325 lines. Branch guide: scope, the file-by-file map of the module, the decisions, the evidence, the known gaps, and the integration notes
research-log.md	374 lines. Per-decision reasoning with official sources - the streaming contracts for Ollama, Groq and Gemini, why nanoseconds and seconds are converted where they are, and what is deliberately not measured

Connected folders

- `src/modules/benchmark/` - the module documented here.
- `docs/internal/` - `benchmark-api.md` is the wire contract these documents refer to.
- `docs/local-model-setup.md` - the setup guide, same work package.
- `evidence/benchmark/` - the artefacts the research log cites.

Referenced from

These folders point here. Each link below resolves in both directions.

- `scripts/benchmark/`
- `src/modules/benchmark/infrastructure/providers/`

docs/dashboard/ - dashboard module documentation

Work package: product UI and benchmark dashboard / integration lead.

What this folder is

One document describing the dashboard as a module: the user journey, the design decisions, and - usefully - the dependency requests it made of the other work packages.

What happened here

The document was not rewritten, and reading it now is the clearest single view of what changed in this project, because several of its open requests have since been closed:

- *"POST /api/v1/workloads` currently echoes `temp--id` and does not persist"** - fixed; the route writes real rows.
- *"the dashboard detects the placeholder ids and accepts an existing database UUID instead"* - that fallback path is gone, along with the UUID field.
- *"A `user\_id`-less benchmark history needs sessions"* - sessions exist.

The document also records the two decisions that survived unchanged and are worth keeping: one typed gateway for every request, and client validation that mirrors rather than replaces server validation.

Files

File	What it is
README.md	70 lines. The four-step user journey, the design decisions (one typed gateway, module-scoped styling under .epd, mirrored validation, accessibility), and the dependency requests to the other work packages

Connected folders

- src/components/dashboard/ - the components described.
- src/app/dashboard/ - the route and its stylesheet.
- docs/internal/ - benchmark-api.md, the contract the dashboard is written against.

Referenced from

These folders point here. Each link below resolves in both directions.

- docs/
- src/components/

docs/internal/ - maintainer documentation

Work package: integration and API surface, with `benchmark-api.md` belonging to the benchmark layer.

What this folder is

Documentation aimed at people working *on* the codebase rather than using it: architecture, database, the wire contract, and how the team works.

What happened here

`README.md` was updated to describe the current document set.

`benchmark-api.md` remains the authoritative wire contract - the dashboard's typed gateway names it in its own header as the thing it is written against, including the awkward cases: "all providers failed but the run is still evidence", and `persisted: false`.

`database.md` is **stale and known to be stale**. Line 147 still describes users owning "devices", a table that no longer exists, and the document does not mention four things that do: `shared_findings`, `users.session_id`, `benchmark_results.warmup`, and `readiness_scores.privacy_class`. It is listed in section 6 of `00-PROJECT-RECORD.md` rather than silently patched, because a documentation gap that is written down is a task and one that is not is a trap.

Files

File	What it is
<code>architecture.md</code>	401 lines. Hexagonal architecture, module boundaries, data flow, schema notes
<code>benchmark-api.md</code>	317 lines. Every endpoint of the benchmark layer with exact request and response shapes, every status code, and the eight error codes. Includes why ownership failures return 403 rather than 404
<code>database.md</code>	167 lines. Local versus shared database, migrations, seeding, conventions. Currently stale - see above
<code>team-workflow.md</code>	367 lines. Branching, issue templates, PR process and size limits, review checklist, conventional commits, daily standup, hotfix procedure
<code>project-plan.md</code>	28 lines. Internal milestones and evaluation notes
<code>Documentation.md</code>	45 lines. An earlier overview of the framework
<code>README.md</code>	27 lines. Index of this folder and pointers to the public-facing documents

Connected folders

- `prisma/` - the schema `database.md` describes.
- `src/app/api/v1/` - the endpoints `benchmark-api.md` specifies.
- `docs/benchmark/` - the module guide that refers back to the wire contract.

Referenced from

These folders point here. Each link below resolves in both directions.

- docs/
- docs/dashboard/
- scripts/db/
- src/
- src/app/api/
- src/app/api/v1/benchmarks/
- src/modules/

docs/vision-benchmark/ - vision module documentation

Work package: vision benchmark.

What this folder is

Seven documents covering the vision workload end to end: what the dataset is, what the workload specifies, how to execute it against a provider, what the test cases are, and how to deliver it.

What happened here

Not rewritten. Two things in here now describe behaviour that has changed underneath them and should be read with that in mind:

- The **dataset card** describes the seven-label construction dataset as the workload's dataset. That is still the reference dataset, but the module no longer requires seven labels - a user can bring their own classes from the browser.
- The **workload specification** fixes the prompt. The prompt is now built from whatever label set the dataset declares (`buildVisionPrompt(labels)`), which a deliberate failure run proved was necessary: with the fixed prompt and a custom two-class dataset the result was 0% accuracy and 100% invalid output.

Files

File	What it is
dataset-card.md	89 lines. Identity, provenance, licence, generation method, privacy checks
workload-specification.md	126 lines. What the workload sends, what it expects back, the thresholds it is graded against
provider-execution.md	106 lines. Prerequisites and how to run against Ollama or Gemini
test-cases.md	60 lines. The 48 documented cases in the automated suite
evidence-report.md	50 lines. Automated verification summary
live-modification-demo.md	51 lines. A worked demonstration of changing the workload and seeing the measurement move
pr-delivery.md	47 lines. Suggested pull-request title and body

Connected folders

- `src/modules/vision-benchmark/` - the module.
- `datasets/vision-benchmark/` - the dataset the card describes.
- `evidence/vision-benchmark/` - the runs, including the one that fails its gate.
- `scripts/vision-benchmark/` - the commands `provider-execution.md` documents.

Referenced from

These folders point here. Each link below resolves in both directions.

- `datasets/`
- `docs/`

- src/app/vision-benchmark/